Predicting Corporate Bond YTM Changes

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1 Objective

The objective of this project is to predict corporate bond yield to maturity (YTM) changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators. By applying various machine learning regression models such as ElasticNet, Boosting, and Multi-layer Perceptron, we aim to improve prediction accuracy compared to traditional linear regression models. In addition, we used the change direction of YTM movements as a classification target since there are many outliers in the YTM change values that may affect regression model performance. We applied classification models such as Logistic Regression, ElasticNet Classifier, and Boosting Classifier to predict the direction of YTM changes.

2 Methodology

2.1 Data Description

We utilized data from Wharton Research Data Services (WRDS) to construct our dataset. The data frequency is monthly, and the data range is from July 2002 to February 2025. We have a total of 50 features from four categories: bond-level features, firm-level features, macroeconomic features, and industry features. The details of the predictors are listed in the appendix.

The target variable is the corporate bond yield to maturity (YTM) change, defined as the difference in YTM over a one-month horizon:

$$\Delta \text{YTM}_i = \text{YTM}_{i,t} - \text{YTM}_{i,t-1} \tag{1}$$

where $\text{YTM}_{i,t}$ is the yield to maturity of bond i at the end of month t.

For classification, the target variable is defined as:

$$\text{YTM\_Direction}_i = \begin{cases} \text{up}, & \text{if } \Delta\text{YTM}_i > 1e^{-6} \\ \text{neutral}, & \text{if } \Delta\text{YTM}_i \approx 0 \\ \text{down}, & \text{if } \Delta\text{YTM}_i < -1e^{-6} \end{cases} \tag{2}$$

The total number of observations after merging all datasets is approximately 854,769, covering 12,117 unique bonds over the sample period.

The data descriptive statistics of the yield to maturity (YTM) changes and class distribution are as follows:

	count	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**
Value	854769.00	-0.000005	0.01444	-0.9808	-0.00184	-0.00003	0.00171	0.97903

Table 1: Descriptive statistics

	down	**up**	**neutral**
Count	435398	399205	20166

Table 2: Class distribution

2.2 Feature Construction

After preparing the bond-level, firm-level, sector-level, and macroeconomic datasets, we align all information at a monthly frequency and construct the final panel used for modeling.

1. **Merging Procedure** We combine the datasets using index columns: CUSIP (bond identifier), PERMNO (firm identifier), and Date (end-of-month date).

2. **Missing-Value Handling:** Missing values are handled based on variable type:

   - **Categorical Variables:** We use cross-sectionally monthly median to impute missing values, preserving the categorical distribution across firms each month. For example: credit rating dummies (AAA. . . D), rating transition indicators, and Compustat data-status flag (costat).

   - **Numeric Variables:** We also use cross-sectionally monthly median to impute missing values for numeric variables, the remaining NaNs are set to be zero.

3. **Normalization** We apply two layers of normalization. First, all bond-level and firm-level variables are rank-normalized within each month. The variables are mapped to [-1, 1] using:

$$\text{scaled\_rank} = 2 \times \frac{\text{rank}}{N} - 1 \tag{3}$$

   where rank is the cross-sectional rank of the variable for that month, and N is the total number of non-missing observations that month. Features in this category include stock features, fundamentals, derived ratios, growth variables, and bond-level numeric predictors. For binary variables (e.g., upgrade/downgrade), ranks are assigned such that 1 maps to 1 and 0 maps to -1.

   Second, macroeconomic variables are normalized using z-score normalization based on training set statistics in each rolling window to avoid data leakage as well as keep their time-series properties.

$$\text{normalized\_value} = \frac{\text{value} - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{4}$$

   where μ_{train} and σ_{train} are the mean and standard deviation of the macro variable computed from the training set only.

2.3 Data Splitting

We used a rolling window forecasting approach. At each year, we used a time-series split to create training, validation, and test sets. The training set consisted of 10 years of data, the validation set and test set each consisted of 1 year of data. At each year, we tuned hyperparameters based on training set and validation set, then forecast target variables in the test set. When tuning is not involved or cross-validation is adopted, the validation set was combined with the training set to train the final model. For example, to predict YTM changes in 2013, we used data from 2002 to 2011 as the training set, data in 2012 as the validation set, and data in 2013 as the test set.

2.4 Modeling Framework

2.4.1 Regression Models

We implemented and compared the following machine learning models for the regression task:

Model	**Description**
Linear Regression	Baseline model used for comparison.
ElasticNet Regression	Combination of L1 and L2 regularization, with hyperparameters α and $l1\_ratio$ tuned via 10-fold cross-validation.
Boosting Regressor (LightGBM)	Gradient boosting model built from ensembles of weak learners with early stopping. Hyperparameters such as num_leaves and learning_rate are tuned using grid search.
Stacking Regressor	Ensemble of multiple base models, using a linear regression meta-learner.
Multi-layer Perceptron	Feedforward neural network.

The MLP is implemented as a small fully connected feed-forward network with ReLU activation functions and a single linear output node, trained with mean squared error loss and the Adam optimizer. For each rolling window, we re-estimate the MLP on the corresponding training period and evaluate it on the test year, using an internal hold-out split within the training window for early stopping. Thus, the MLP is directly comparable to the other regression models in terms of the rolling-window evaluation design, although it is considerably more computationally intensive and we keep the network architecture relatively small to control training time. The MLP is not part of the stacked regressor due to its separate training procedure and high computational cost.

2.4.2 Classification Models

For classification task, we implemented and compared the following models:

Model	**Description**
Logistic Regression	Baseline model used for comparison.
ElasticNet Classifier	Combination of L1 and L2 regularization. We tuned alpha and l1_ratio hyperparameters using 10-fold cross-validation.
Boosting Classifier (LightGBM)	Ensemble of weak learners. We tuned num_leaves and learning_rate hyperparameters using grid search.
Stacking Classifier	Ensemble of multiple models using logistic regression as meta-model

2.5 Model Evaluation

We used a rolling window evaluation approach and calculated performance metrics on the test set for each year from 2012 to 2024. Then, the metrics were averaged to assess overall model performance. For regression models, performance metrics included: MSE, MAE, MedAE, and R^2 For classification models, performance metrics included: Accuracy, Precision, and Recall

3 Results and Discussion

3.1 Regression Results

The Elastic Net Regression model does not significantly outperform the Linear Regression benchmark, indicating that regularization alone provides only limited gains in this setting. The LGBM Regressor and the Stacking Regressor achieve slightly lower MSE than the pure linear model, but their average out-of-sample R^2 remains close to zero (and sometimes slightly negative), suggesting only modest incremental predictability from these nonlinear ensembles. The tuned multilayer perceptron (MLP) achieves the highest average out-of-sample $R^2 \approx 0.064$ across the rolling windows, indicating that the nonlinear network is able to capture some additional time-series variation in YTM changes. However, this comes at the cost of slightly higher MSE/MAE compared with the best tree-based and stacked models, and the overall magnitude of the R^2 remains modest. This suggests that, given our current feature set and network size, a feed-forward neural network provides only limited improvement in predictive accuracy over simpler linear and ensemble benchmarks.

Model	MSE	MAE	MedAE	R^2
Linear Regression	0.000094	0.003226	0.002065	0.001753
ElasticNet Regression	0.000097	0.003107	0.001900	-0.019272
Boosting Regressor	0.000089	0.003069	0.001620	-0.109083
Stacking Regressor	0.000086	0.002959	0.001591	-0.008823
Multilayer Perceptron	0.000105	0.003522	0.002168	0.064329

3.2 Classification Results

In classification tasks, the Boosting Classifier (LightGBM) and Stacking Classifier outperformed the logistic models.

Model	Accuracy	Precision	Recall
Logistic Regression	0.687858	0.687190	0.687858
ElasticNet Classifier	0.687915	0.686969	0.687915
Boosting Classifier (LightGBM)	0.713470	0.719949	0.713470
Stacking Classifier	0.713615	0.720544	0.713615

4 Conclusion

In this project, we explored various machine learning models to predict corporate bond yield to maturity changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators. Our findings indicate that ensemble methods like LightGBM provide clear gains in the classification task, where they outperform traditional linear models in terms of accuracy, precision, and recall. For the regression task, they also offer modest improvements over linear benchmarks. The MLP model achieves worst MSE suggesting we do not need complex neural networks for this problem given the current feature set.

A Appendix

A.1 Implementation Details

1. **Software and Libraries**

    - CPU: MacBook Pro (Apple M4 Pro)
    - Programming Language: Python 3.13
    - Libraries: pandas, numpy, scikit-learn, lightgbm, tensorflow/keras
    - Code Repository (GitHub)

2. **Computation Time For Each Model (Approximate)**

Model	**Training/Prediction**
Linear Regression	3.7 seconds
Elastic Net Regression	9.2 seconds
Boosting Regressor	2 minutes 15 seconds
Stacking Regressor	4 minutes 50 seconds
Logistic Regression	57 seconds
Elastic Net Classification	635 minutess
Boosting Classifier	10 minutes
Stacking Classifier	19 minutes 7.4 seconds
Multilayer Perceptron	38 minutes 48 seconds

3. **AI Tools Used:** We used ChatGPT for drafting and refining report text, and GitHub Copilot for code suggestions and completions.

A.2 Predictor List

Table 3: Comprehensive Description of All Predictors

Field Name	**Description**
A. Key Identifier Columns	
CUSIP	Committee on Uniform Securities Identification Procedures. Unique 9-digit identifier for each bond.
PERMNO	Permanent Number. Unique firm-level identifier.
Date	End of Month Date.
B. Bond-level Features	
`tmt`	The remaining time to maturity of the bond in years.

Table 3 – Continued from previous page

Field Name	Description
coupon	The annual coupon rate of the bond.
t_spread	The weighted average bid-ask spread of the bond which measures liquidity.
return_eom	Bond return of last month.
Credit Ratings	One-hot encoded variables for ratings including AAA, AA, A, BBB, BB, B, CCC, CC, C, and D.
upgrade	Binary variables indicating if the bond was upgraded in the last month.
downgrade	Binary variables indicating if the bond was downgraded in the last month.
C. Firm-level Features: Stock Market	
ret	Monthly stock return accumulated from daily returns.
vol	Monthly return volatility, calculated as the standard deviation of daily log returns.
dvol	Dollar trading volume, calculated as price multiplied by shares traded.
turnover	Share turnover ratio, defined as shares traded divided by shares outstanding.
bidask	Bid-ask spread based on daily closing bid and ask quotes, measuring liquidity cost.
numtrades	Total number of trades executed in the month.
mktcap	Month-end market capitalization (Price $\times$ Shares Outstanding).
price_mean	The average daily closing stock price over the month.

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Field Name	Description
D. Firm-level Features: Accounting-Based Derived Indicators	
`log_atq`	Natural logarithm of total assets, used as a proxy for firm size.
`lev_total`	Total leverage ratio, calculated as total liabilities divided by total assets.
`equity_ratio`	Equity-to-assets ratio, calculated as total common equity divided by total assets.
`roa`	Return on Assets (ROA), defined as net income divided by total assets.
`profit_margin`	Net profit margin, calculated as net income divided by total revenue.
`int_coverage`	Interest coverage ratio, defined as operating income divided by interest expense.
`mkt_cap`	Accounting-based market capitalization.
`market_to_book`	Market-to-book ratio, calculated as market capitalization divided by book equity.
E. Firm-level Features: Growth Indicators (Quarter-over-Quarter)	
`atq_growth`	Quarter-over-quarter growth rate of total assets.
`revtq_growth`	Quarter-over-quarter growth rate of total revenue.
`niq_growth`	Quarter-over-quarter growth rate of net income.
F. Industry Features	

Table 3 – Continued from previous page

Field Name	Description
`price`	The monthly closing price of the sector ETF corresponding to the firm's Global Industry Classification Standard (GICS) sector.
`return`	The monthly return of the sector ETF, used as a proxy for sector-specific market performance and shocks.
G. Macroeconomic Features	
`sp500`	Monthly level of the S&P 500 index, capturing overall equity market conditions and risk appetite.
`ir3m`	Monthly average 3-month Treasury yield, representing short-term risk-free rates and monetary policy stance.
`ir10y`	Monthly average 10-year Treasury yield, reflecting long-term discount rates relevant for bond valuation.
`vix`	Monthly average VIX index, measuring market-implied volatility and broad risk aversion.
`gdp`	Real GDP level or growth, aligned to each month, summarizing aggregate economic activity.
`cpi`	Consumer Price Index (level or change), measuring inflation conditions relevant for nominal yields and real required returns.